

SARBESH BANJARA

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WORK EXPERIENCE

Graduate Research Assistant, The University of Alabama, Tuscaloosa, Alabama

Jan 2025- present

- Built a scalable, automated framework for building level damage assessment from 360-degree video, replacing manual inspection workflows
- Established a methodology to propagate spatial and sensing uncertainty into structural damage interpretation.
- Enabled reliable use of large-scale public footprint datasets by quantifying and correcting spatial inconsistencies.
- Data driven analysis of tornado induced building damage using information theoretic methods and machine learning to identify damage patterns and inform mitigation strategies
- Evaluated computer vision and AI models for tornado damage assessment by testing their accuracy across many experiments and datasets

Government of Nepal

Senior Civil Engineer

Jan 2023 – Dec 2024

Civil Engineer

Aug 2018 – Jan 2023

o **Minister of Internal Affairs and Law, Bagmati Province**

- Provide administrative and technical support to the Minister of Internal Affairs and Law.
- Actively involved in the development and implementation of disaster management plans for floods and landslides during the monsoon season, including but not limited to early warning systems and evacuation procedures, relief efforts, and emergency preparedness.
- Coordinate with other government agencies and stakeholders on planning, design, and construction of construction projects.

o **Division Chief**, Building Permit and Standard Implementation Division, Hetauda Sub-Metropolitan City

- Oversaw the issuance of building permits and ensured compliance with all relevant building codes and standards.
- Periodically led teams of engineers and inspectors in conducting inspections of construction sites and buildings to identify and address any safety concerns or code violations.

o **Division Chief**, Plan Implementation and Public Works Division, Hetauda Sub-Metropolitan City

- Formulation and implementation of annual and periodic plans of Hetauda Sub-Metropolitan City related to Infrastructure development.
- Periodically led teams of engineers and inspectors in conducting inspections of construction sites and identifying and addressing any safety concerns or code violations related to public infrastructure development works conducted by Hetauda Sub-Metropolitan City.

o **Civil Engineer**, Ministry of Physical Infrastructure Development, Bagmati Province

- Worked on projects related to physical infrastructure development in Bagmati Province, including roads, bridges, and irrigation systems.
- Coordinated with other government agencies and stakeholders to ensure the smooth implementation of projects.

o **Civil Engineer**, Ministry of Water Supply, Government of Nepal

- Worked on diverse projects related to water supply development and management in the Bagmati Province
- Developed and implemented water supply plans and programs at the national and local levels.
- Monitored and evaluated the performance of water supply systems in rural areas of Bagmati Province near Hetauda Sub-Metropolitan City

Project Civil Engineer, Riddhi Siddhi Cement Pvt. Ltd., Hetauda, Bagmati Province

Apr 2018 – Aug 2018

- o Performed Quality Control (QC) of fresh concrete, mortar, and cement silos by ensuring compliance with all relevant standards and specifications.
- o Monitored the progress of all construction activities to ensure they were completed in time and within budget, and that they met all relevant safety and quality standards.
- o Performed preliminary survey of building design to assess their suitability for construction, considering the factors such as soil conditions, topography, and access.

Civil Engineer, Prithik Engineering Consultancy, Hetauda, Bagmati Province,

Jan 2018 – Apr 2018

- o Performed valuation of land and buildings by analyzing comparable sales data, market trends, and property characteristics
- o Designed and prepared construction drawings for residential, commercial, and industrial structures, adhering to all relevant codes and standards.

- o Prepared Detailed Project Reports (DPRs) for roads, bridges, and buildings, including sections on scope, goals, budget, timeline, and risks.

Surveyor, Kathmandu-Terai Fast Expressway Track (National Pride Project), Bagmati Province *Dec 2017 – Jan 2018*

- o Performed collaboration with the preliminary survey team to collect data on road conditions and geometry.
- o Used surveying equipment, including but not limited to Total Stations, Theodolites, GPS, and Levels to perform alignment and cross-section survey.
- o Analyzed survey data to identify potential road hazards and suggested recommendations for improvement.
- o Performed quality control (QC) of the obtained data to ensure the data was collected accurately and efficiently.

ACADEMIC QUALIFICATION

- **PhD in Civil Engineering**, The University of Alabama, Tuscaloosa, Alabama *Jan 2025 – present*
- **MS in Construction Management**, United Technical College, Pokhara University, Pokhara, Nepal, *Dec 2022 – Nov 2024*
- **Master's in Public Administration**, Tribhuvan University, Kathmandu, Nepal *Apr 2017-Nov 2019*
- **BE in Civil Engineering**, Tribhuvan University, Kathmandu, Nepal *Nov 2013-Dec 2017*

Key Professional Services

- Successfully led a team of professionals in drafting a new planning standard for group housing projects, which was later converted into municipal legislation of Hetauda sub-metropolitan city, Nepal, and has served as a benchmark guideline for group housing projects in Hetauda sub- metropolitan city.
- Actively involved in the development of a new land use classification guideline of Hetauda Sub-metropolitan city, which was later converted into municipal legislation, and it has guided the development of plans, policy, and budget allocation for the city.
- Invited as a subject matter expert to deliver a technical session on building permit systems and standards implementation for the Bakaiya rural municipality, which supported the municipality in initiating its building permit system and regulatory enforcement process.
- Designed and supervised hundreds of civil infrastructure projects, ensuring compliance with codes, quality standards, and project schedules.
- Supervised and implemented building bylaws and design standards across hundreds of residential, commercial and government building projects.

Reviewing and Judging

- Contributed as a peer reviewer for 3 (Three) Engineering, Construction and Architectural Management Journal papers, supporting the integrity and quality control of academic research through critical evaluation of manuscripts
- Contributed as a peer reviewer for Journal of Environmental Sustainability (Q2) (2 Reviews, One manuscript), supporting integrity and quality control of academic research through critical evaluation of manuscripts
- Contributed as a peer reviewer for Journal Journal of UTEC Engineering Management supporting integrity and quality control of academic research through critical evaluation of manuscripts
- Peer reviewed multiple research abstracts for the NSF NHERI Graduate Student Council Mini Conference, assessing technical quality, methodological rigor, and relevance to natural hazards engineering.
- Invited to serve as a panel judge for the ASCE Gulf Coast student symposium paper and presentation competition alongside distinguished professionals, recognizing expertise in evaluating technical rigor, research contribution, and presentation quality.
- Served as a judge at the Undergraduate Research and Creative Activity Conference, The University of Alabama assessing student research projects based on technical quality, originality, and presentation

Research Milestones

- Invited by the National Institute of Standards and Technology (NIST), the premier U.S. federal research agency, and delivered a seminar to a distinguished professional audience in Summer 2026 on emerging geospatial and image-based methods for advancing rapid and reliable post-disaster building assessment, a critical area for strengthening U.S. disaster response and resilience.
- Conducted analysis of 2011 Joplin tornado damage that supported ASCE technical committee discussions on wind-speed criteria for glazing protection, with potential contributions to future ASCE 7 updates and improved tornado-resistant design practices in the United States.
- Designed and supervised hundreds of civil infrastructure projects, ensuring compliance with codes, quality standards, and project schedules.

□ Publications

- Banjara, S., & Shrestha, S. (2025). Analysis of the Cost-Effectiveness of Retrofitting Using the Benefit-Cost Ratio and the Breakeven Point Analysis. Journal of UTEC Engineering Management, 3(1), 16–26. <https://doi.org/10.3126/juem.v3i1.84513>
- Umeike, Robinson, Thang Dao, Shane Crawford, John van de Lindt, Blythe Johnston, Wanting Lisa Wang, Trung Do, Ajibola Mofikoya, Sarbesh Banjara, and Cuong Pham. "ASTRA: architectural insights for post-tornado damage recognition." Expert Systems with Applications (2026): 133915. <https://doi.org/10.1016/j.eswa.2026.133915>
- Banjara, S., Crawford, P.S., "Evaluating Building Footprint Data Quality for Reliable Post-Disaster Image Assessment, 2026" (Accepted, 11th forensic engineering congress)

□ Media Coverage

- Research Featured in Nepal TV World
 - Topic: Seismic Retrofitting, Long-term Cost-Effective Solution: Study
 - Coverage of my research on seismic retrofitting as a long-term cost-effective strategy
 - Highlights real-world application in earthquake-prone regions.
 - Link: <https://nepalworld.com/seismic-retrofitting-long-term-cost-effective-solution-study/>

TRAININGS

- Obtained training in 'Climate Resilient Infrastructure Development' hosted by Town Development Fund.
- Attended a training program on the 'Electronic Building Permit System' organized by UNDP and the Ministry of Federal Affairs and General Administration
- Participated in 'Earthquake Resistance Development' organized by the National Reconstruction Authority.
- Completed training on 'Accessible Infrastructure Design' organized by the National Federation of the Disabled Nepal.
- Received training on 'Electronic Government Procurement' hosted by Public Procurement Monitoring Office.
- Coordinated the Civil Exposition organized by CESS-Nepal every year for three years during my undergraduate studies.

SOFTWARE

ETABS	Civil 3-D	MS-Office	AutoCAD	Land Development
SLOPE/W	SAP2000	MATLAB	Python	Arc GIS

CERTIFICATIONS / MEMBERSHIPS/ HONORS

- FE passed, NCEES, Michigan
- Student Member, American Society of Civil Engineers
- Student Member, American Concrete Institute
- Member, Natural Hazards Engineering Research Infrastructure, Graduate Students Council
- Nepal Engineering Association (NEA) Certified
- Nepal Engineering Council (NEC) registered
- Acted as Committee Member of Class-D Contractor Licensing Committee
- Acted as a representative in various committees of the sub-metropolitan city, of which the head of the plan implementation and public works division is an ex officio member
- Acted as an expert member of the recruitment committee of Hetauda Sub-Metropolitan City for the recruitment of Engineers and Overseers.
- Selected as a government Civil Engineer in the Public Service Commission (PSC) exam in 2018.
- Undergraduate merit-based scholarship (selected within the top 108 students among 13,000+ candidates in the Engineering Entrance Exam conducted by Tribhuvan University)
- Intermediate-level merit-based scholarship